So, you can check the index and look for the name Bloom. Being in lexicographical order, you could go through it from the start till you catch the word you are looking for: shouldn’t be too long, with Bloom. You wouldn’t be that lucky with terms hashing or, even worse, tree, which will be towards the end of the index.

That’s why, naturally, we do lookups in sorted lists by subconsciously using binary search[4]: with a phone book, you open it at a random page around the middle (or closer to beginning or end, if you have an idea where the name you are looking for might stand), then look the first letter of the first word on that page, and jump either before or after current page depending on what you are looking for. For example, if you are still looking for Bloom, and you open a phone book on a page where the first surname is Kurtz, then you know you can discard every page after that, and look only at the pages before. You randomly open another page (on the left of the one with Kurtz) and the last name on that page is Barrow: then you know Bloom will be in a page after the one with Barrow and before the one with Kurtz.

Going back to our problem with the contact list, one approach could be sorting your contacts and searching them using binary search.

Table 4.2 Using Sorted Arrays as Dictionaries

Operation	Running Time	Extra Memory
`Creating the structure`	O(n*log(n))	O(n)